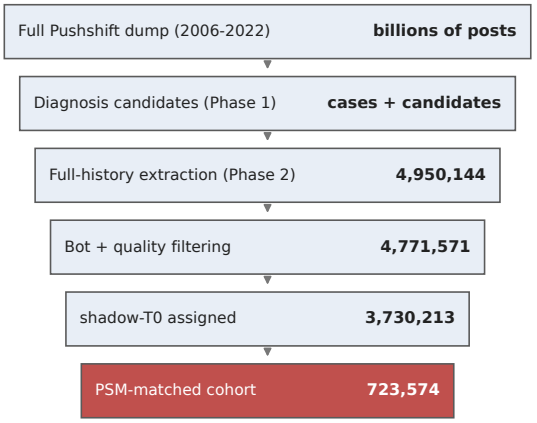
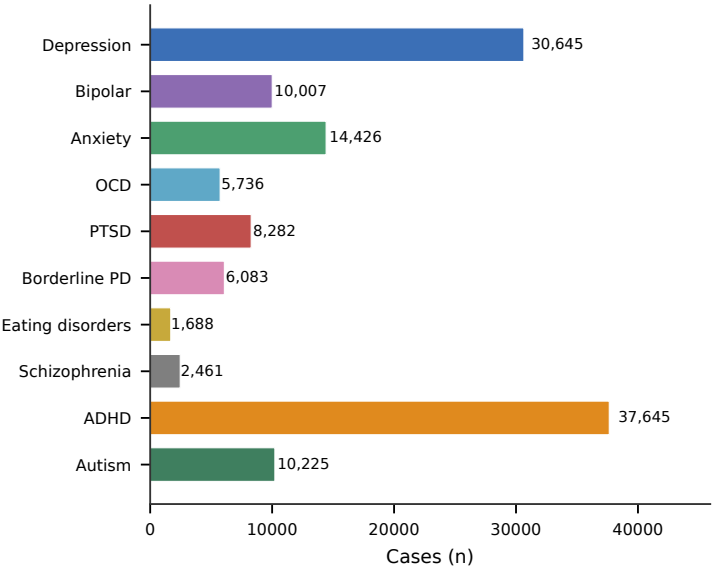
a Cohort construction



cases 128,003 · matched comparisons 595,571
1:4.65 matching, all SMD < 0.1

b Case composition by disorder



c What digital phenotyping can and cannot do

Capability	Evidence	Verdict
Case-control discrimination	AUROC 0.84-0.93	Robust
Language-independent detection	dAUROC 0.006	Robust
Temporal reproducibility	t0-surge 0.11	Robust
Single-family detection	0.60-0.83	Partial
Disorder separation	macro AUROC 0.72	Moderate
Discrete clustering	ARI 0.088	Weak
Individual early warning	sparse pre-t0 CPs	Unsupported
Causal inference	DID + placebo	Inconclusive
Population volatility	detrended r 0.47	Promising
Population severity trend	no robust signal	Not supp.

d Scale-dependent validity

Individual detection — strong — **Robust**

Disorder vs disorder — moderate — **Moderate/weak**

Latent structure — shared / dimensional — **Dimensional**

Individual early warning — not supported — **Unsupported**

Population monitoring — behavioural only — **Behavioural only**

PSM layer: 128,003 cases (127,198 labelled across 10 disorders, 805 unlabelled) and 595,571 matched comparisons.
Funnel tallies from the preprocessing pipeline. Panels b-d computed from NAS result files.